

TRANSSERIES / ANALYTIC INVERSION / COUNTABLE ACTION FAMILIES

# The Logarithmic Critical Endpoint

## Convergent Lambert Charts, All-Order Sector Laws, and Sharp Action-Cutoff Corrections

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### Abstract

For the positive countable-action equation

$$U_\beta(q) = \beta \sum_{j \geq 1} a_j q^j e^{j U_\beta(q)}, \quad a_j = a j^{-3} \quad \text{outside a finite set,}$$

the critical inverse is neither an analytic square-root fold nor the fractional Hahn chart associated with a nonintegral power tail. We derive its logarithmic normal form and construct an exact, convergent two-variable analytic chart over the negative real branch of Lambert's  $W$  function. An exact Lambert normalization also yields an arbitrary-order, uniformly additive inverse-logarithmic expansion of the sector coefficients, with explicit Gaussian–logarithmic Fourier kernels. We then obtain a uniform finite-coupling refinement of the action-cutoff profile, including its  $1/\sqrt{\log n}$  and  $(\log \log n)/\log n$  corrections. Two scales separate: fluctuations have order  $\sqrt{n \log n}$ , whereas preserving the  $n$ th sector requires and, in the critical window, is equivalent to retaining  $M \gg \sqrt{n}$  actions. A convergent implicit chart gives all inverse-logarithmic orders of the finite-action critical-point drift and explains the same cutoff law from moving analytic folds. A final theorem extends the leading scale separation to logarithmically weighted cubic tails, including the borderline  $j^{-3}/\log j$  case. Conventional proofs, exact algebraic checks, and reproducible numerical diagnostics are supplied.

**Scope and research status.** The fixed upper endpoint of the earlier countable-action article's Research Question 4 is treated here; the joint limit in which its exponent approaches 2 with  $n$  is not. The leading Gaussian/extreme-value separation has classical predecessors, notably Janson's Example 18.29. The contributions developed here are the explicit charts, correction formulas, and their proof-level connection for this model, not a claim to have discovered that classical mechanism. Publication novelty has not been established by an exhaustive literature review. No Lean formalization or interval-certified numerical implementation is claimed.

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# 1 The research target and its precise boundary

## 1.1 Why this endpoint is not obtained by substitution

The repository's transseries program develops polynomial-logarithmic calculus, inversion around exactly invertible cores, and certified residual transport. Recent packages also study analytic moving folds and countable exponential feedback [6, 7]. The preceding research article *Beyond Finite-Action Folds* treats an eventually exact tail  $a_j = aj^{-1-\alpha}$  with  $1 < \alpha < 2$  [8]. At criticality its normal form begins with  $\Gamma(-\alpha)x^\alpha$ ; the critical chart, the sector fluctuation scale, and the action budget are organized by this nonintegral exponent.

The upper endpoint is qualitatively different. The pole of  $\Gamma(-\alpha)$  at  $\alpha = 2$  cancels a divergent analytic quadratic coefficient. The surviving expression is  $x^2 \log(1/x)$ , not a finite multiple of  $x^2$ . Its inverse needs a logarithmic correction, while its sector coefficients lie in the infinite-variance Gaussian domain. The largest relevant individual actions are now smaller than the fluctuation scale. Substituting  $\alpha = 2$  into the nonintegral formulas loses all three features.

Research Question 4 of the earlier article explicitly asks for endpoint logarithmic charts and cutoff laws. This paper treats the *fixed upper endpoint*, with finite-prefix perturbations. It also supplies an asymptotic rate refinement relevant to that article's Question 2, an all-order finite-fold drift chart relevant to Question 7, and a defined slowly varying subclass relevant to Question 1. These are scoped resolutions, not a claim that all of those broader research questions are settled.

*Editorial note (ProveIt, 2026-09-29).* The marginal package [9], filed beside this one, treats the same endpoint (the same model, tail  $aj^{-3}$  outside a finite prefix) and was written independently of this article, also from a library copy of the earlier article. Its core coincides with this article's: the same Lambert- $W_{-1}$  core (this article's  $(v, T, \eta)$ ,  $h$  and  $c$  are its  $(w, t, 2r)$ ,  $b$  and  $A$ ), the same chart coefficients and inverse diagnostics, the same fold functions, and the same critical cutoff correction, which it states with the sharper error  $O(H_n^{-3/2})$ . Its sector constants  $\kappa_1, \kappa_2$  are  $g_1, g_2$  of Corollary 5.3 rewritten in the normalization  $H_n = \log B_n$  in place of  $\ell_n = \log b_n + h$ ; in particular  $\kappa_1 = g_1 - h/2 = (2 - \gamma - \log 2 - 2b)/4$ . It is one theorem with two independent proofs. The Poisson process, the log-weighted tails, the finite-fold local limit and the exponentiation criterion are only here; its explicit contraction radii, three-generator logarithmic grid, two-term minimal budget and finite retention inequality are only there.

## 1.2 Classical background and the contribution developed here

The substitution  $t = qe^U$  gives  $t = q \exp(\beta A(t))$ , a classical implicit function schema. Lagrange inversion and Fourier local-limit methods are standard tools [3, 4]. In particular, Janson's Example 18.29 treats cubic power-law weights in the infinite-variance Gaussian case and explains why conditioning does not change the leading extreme-value behavior [4]. The leading scale separation in this paper must be read in that context, not as an unresolved general probabilistic phenomenon. The broader distinction between soft and hard truncations also has an established literature [5].

What is developed explicitly here is a linked analytic and computational calculus for the action model. The convergent Lambert chart retains the entire analytic remainder instead of stopping at the leading singular term. The inverse-logarithmic sector expansion fixes an exact normalization and gives a formula for every correction. The refined cutoff law keeps finite coupling, omitted mean, omitted mass, and retained variance separate. The finite-fold chart identifies a single convergent function whose Taylor coefficients determine the full inverse-logarithmic drift. The proofs are included rather than delegated to an unreviewed predecessor.

| Result family                                  | Status and contribution in this article                                                                    |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Implicit substitution and Poisson conditioning | Classical identities, proved here in the exact normalization used later.                                   |
| Leading Gaussian and extreme scales            | Classical mechanism; specialized and proved directly for action counts.                                    |
| Lambert–analytic critical chart                | Exact convergent normal-form construction, explicit coefficients, and a geometric truncation bound.        |
| All-order sector corrections                   | Explicit Gaussian–logarithmic kernels with uniform additive remainders at every inverse-logarithmic order. |
| Refined cutoff law                             | Uniform finite-coupling correction with a stated error order, plus an equivalent action-budget criterion.  |
| Moving finite folds                            | Convergent drift and curvature charts with finite- $M$ errors and exponentiation conditions.               |

### 1.3 Reading conventions

All limits have  $n \rightarrow \infty$ ,  $M \rightarrow \infty$ , or  $\varepsilon \downarrow 0$ , as specified. Logarithms of positive numbers are real. Complex logarithms use the principal determination unless a continuation is explicitly stated. Uniformity in a parameter on a compact set means that the implicit constants may depend on that compact set and on the fixed weight family, but not on  $n$ ,  $M$ , or the varying parameter. “All orders” means one proved estimate for each fixed truncation order; it does not assert convergence or optimal truncation of an asymptotic series.

## 2 The positive action model and exact coefficient identities

Fix  $a > 0$  and a polynomial  $P(t) = \sum_{j=1}^J p_j t^j$  such that

$$A(t) = a \operatorname{Li}_3(t) + P(t) = \sum_{j \geq 1} a_j t^j, \quad a_j \geq 0, \quad a_1 > 0. \quad (2.1)$$

The coefficients  $p_j$  themselves need not be nonnegative. Define the finite prefix moments and the endpoint constants by

$$p^{[k]} = \sum_{j=1}^J j^k p_j, \quad d_0 = a\zeta(3) + p^{[0]}, \quad d_1 = a\zeta(2) + p^{[1]}, \quad (2.2)$$

$$\beta_c = d_1^{-1}, \quad \rho_\beta = e^{-\beta d_0}, \quad c_\beta = \beta a, \quad h = \frac{3}{2} + \frac{p^{[2]}}{a}. \quad (2.3)$$

Write  $c = c_{\beta_c}$  and  $\rho_c = \rho_{\beta_c}$ . Both  $d_0$  and  $d_1$  are positive and finite. The genuine second moment  $\sum j^2 a_j$  diverges;  $p^{[2]}$  is only a finite-prefix quantity.

For  $\beta > 0$ , let

$$U_\beta(q) = \beta A(qe^{U_\beta(q)}), \quad u_n(\beta) = [q^n]U_\beta(q). \quad (2.4)$$

The coefficient at degree  $n$  is determined by lower coefficients and  $a_1, \dots, a_n$ , so there is a unique formal solution in  $q\mathbb{R}[[q]]$ . The analytic implicit function theorem at  $(q, U) = (0, 0)$  shows that it is convergent near zero. Its coefficients are positive: the summand with  $a_1 > 0$  already generates a positive contribution at every degree.

**Proposition 2.1** (Parametrization and coefficient formula). *With  $t = qe^{U_\beta(q)}$ , one has*

$$q = te^{-\beta A(t)}, \quad U_\beta(q) = \beta A(t), \quad u_n(\beta) = \frac{1}{n} [t^n] e^{n\beta A(t)} \quad (n \geq 1). \quad (2.5)$$

*Proof.* The first two equalities follow by substitution. Lagrange inversion applied to  $t = q \exp(\beta A(t))$  gives

$$[q^n] \beta A(t(q)) = \frac{\beta}{n} [t^{n-1}] A'(t) e^{n\beta A(t)}.$$

Since  $(e^{n\beta A(t)})' = n\beta A'(t) e^{n\beta A(t)}$ , coefficient extraction on the derivative gives the last equality in (2.5). This also proves the identity purely formally.  $\square$

Let  $(Z_j)_{j \geq 1}$  be independent Poisson variables with means  $n\beta a_j$ , and put  $S = S_{n,\beta} = \sum_{j \geq 1} j Z_j$ . The total number of jumps is finite almost surely because  $n\beta \sum a_j < \infty$ ; also  $\mathbb{E}S = n\beta d_1 < \infty$ . Its probability generating function is

$$\mathbb{E}t^S = \exp(n\beta(A(t) - d_0)).$$

Consequently,

$$n\rho_\beta^n u_n(\beta) = \mathbb{P}(S_{n,\beta} = n). \quad (2.6)$$

**Definition 2.2** (Action truncation). For an integer  $M \geq 1$ , replace  $A$  by  $A_M(t) = \sum_{j=1}^M a_j t^j$  in (2.4). Denote the resulting coefficients by  $u_n^{(M)}(\beta)$ , and set

$$Q_M = \sum_{j>M} a_j, \quad S_M = \sum_{j \leq M} j Z_j, \quad R_{n,M}(\beta) = \frac{u_n^{(M)}(\beta)}{u_n(\beta)}.$$

Independence and (2.6) give the exact formula

$$R_{n,M} = e^{-n\beta Q_M} \frac{\mathbb{P}(S_M = n)}{\mathbb{P}(S = n)} = \mathbb{P}(Z_j = 0 \text{ for all } j > M \mid S = n). \quad (2.7)$$

In particular,  $0 \leq R_{n,M} \leq 1$  and it is nondecreasing in  $M$ . This monotonicity will prove the sharp budget criterion without any unproved uniform extrapolation of a compact-parameter asymptotic formula.

### 3 The logarithmic singularity and a convergent Lambert chart

#### 3.1 Deriving the integer-order normal form

The nonintegral polylogarithm expansion in DLMF 25.12.12 excludes positive integer orders in its stated form [1]. At order 3 it is safer to derive the cancellation directly. In a slit neighborhood of 0,

$$\text{Li}_1(e^{-x}) = -\log(1 - e^{-x}) = -\log x + \frac{x}{2} - \frac{x^2}{24} + O(x^4).$$

The difference between  $-\log(1 - e^{-x})$  and  $-\log x$  is analytic for  $|x| < 2\pi$ : the analytic factor  $(1 - e^{-x})/x$  has no zero in this disk. Integrating twice and using the values  $\text{Li}_3(1) = \zeta(3)$  and  $\text{Li}_2(1) = \zeta(2)$  gives

$$\text{Li}_3(e^{-x}) = \zeta(3) - \zeta(2)x + \frac{x^2}{2} \left( \log \frac{1}{x} + \frac{3}{2} \right) + \frac{x^3}{12} - \frac{x^4}{288} + O(x^6). \quad (3.1)$$

The remainder after extracting the displayed logarithm is analytic on  $|x| < 2\pi$ .

Set

$$x = -\log t, \quad \varepsilon = \log(\rho_\beta/q), \quad \delta_\beta = 1 - \beta d_1. \quad (3.2)$$

Equation (2.5) yields the exact local normal form

$$\varepsilon = \delta_\beta x + \frac{c_\beta}{2} x^2 \left( \log \frac{1}{x} + h \right) + x^3 R_\beta(x), \quad U_\beta = \beta d_0 + \varepsilon - x. \quad (3.3)$$

Here  $R_\beta$  is analytic for  $|x| < 2\pi$  and linear in  $\beta$ , with

$$R_\beta(0) = \beta \left( \frac{a}{12} - \frac{p^{[3]}}{6} \right), \quad R'_\beta(0) = \beta \left( -\frac{a}{288} + \frac{p^{[4]}}{24} \right). \quad (3.4)$$

At criticality,  $\delta_{\beta_c} = 0$ . Write  $R = R_{\beta_c}$  and  $r_k = [x^k]R(x)$ .

### 3.2 A general logarithmic inversion theorem

The following theorem isolates the analytic mechanism from the particular polylogarithm. Lambert's  $W_{-1}$  is the real branch with values at most  $-1$  on  $[-e^{-1}, 0]$  [2].

**Theorem 3.1** (Convergent Lambert–analytic chart). *Let  $m \geq 2$  be an integer,  $B > 0$ ,  $h \in \mathbb{R}$ , and let  $R$  be analytic near zero and real on the real axis. Consider the small positive solution of*

$$\varepsilon = Bx^m \left( \log \frac{1}{x} + h \right) + x^{m+1} R(x). \quad (3.5)$$

For all sufficiently small  $\varepsilon > 0$ , define

$$v = -W_{-1} \left( -\frac{me^{-mh}\varepsilon}{B} \right), \quad T = \left( \frac{m\varepsilon}{Bv} \right)^{1/m}, \quad \eta = \frac{m}{v}. \quad (3.6)$$

There is a function  $V(T, \eta)$  holomorphic in a bidisk about  $(0, 0)$ , with  $V(0, 0) = 1$ , such that the solution is exactly

$$x = TV(T, \eta), \quad V^m(1 - \eta \log V) + \frac{\eta}{B} TV^{m+1} R(TV) = 1. \quad (3.7)$$

Moreover,  $V(T, 0) = V(0, \eta) = 1$ . Thus  $V - 1$  is divisible by  $T\eta$  in the local ring of convergent power series. The representation also gives the analytic continuation of this branch in a sufficiently small sector about the positive  $\varepsilon$  axis, with compatible branches in (3.6).

*Proof.* The defining relation for  $W$  gives  $ve^{-v} = me^{-mh}\varepsilon/B$ . It follows that

$$\log(1/T) + h = v/m = 1/\eta, \quad BT^m(\log(1/T) + h) = \varepsilon.$$

Substituting  $x = TV$  into (3.5) and dividing by this last identity gives (3.7). Its left side minus 1 is holomorphic near  $(V, T, \eta) = (1, 0, 0)$ , taking the branch of  $\log V$  near 1. The derivative with respect to  $V$  at that point is  $m \neq 0$ . The analytic implicit function theorem therefore gives a unique  $V$  near 1. At  $T = 0$ ,  $V = 1$  is the locally unique solution; the same holds at  $\eta = 0$ . Divisibility follows from the convergent double Taylor series.

For positive  $\varepsilon$  tending to zero,  $v \rightarrow \infty$ ,  $T \rightarrow 0$  and  $\eta \rightarrow 0$ , so the constructed real  $x$  is positive. The derivative of the right side of (3.5) is

$$Bx^{m-1} \left( m \log \frac{1}{x} + mh - 1 \right) + O(x^m) > 0$$

for sufficiently small  $x > 0$ . Hence it is the unique small positive solution. The branch functions and the bidisk solution admit compatible analytic continuation to a small sector about this positive ray.  $\square$

**Corollary 3.2** (Critical chart for cubic action weights). *In (3.3) at  $\beta = \beta_c$ , put*

$$v = -W_{-1}\left(-\frac{4e^{-2h}\varepsilon}{c}\right), \quad T = 2\sqrt{\frac{\varepsilon}{cv}}, \quad \eta = \frac{2}{v}. \quad (3.8)$$

*Then  $x = TV(T, \eta)$  is a convergent chart and*

$$V(T, \eta) = 1 + v_1(\eta)T + v_2(\eta)T^2 + O(T^3), \quad (3.9)$$

$$v_1(\eta) = -\frac{2\eta r_0}{c(2-\eta)}, \quad (3.10)$$

$$v_2(\eta) = -\frac{2\eta r_1}{c(2-\eta)} + \frac{2\eta^2(10-3\eta)r_0^2}{c^2(2-\eta)^3}. \quad (3.11)$$

*The  $O(T^3)$  is uniform for  $\eta$  in a sufficiently small closed disk. At each fixed order in  $T$ , its coefficient is a rational function of  $\eta$  whose possible poles are at  $\eta = 2$ .*

*Proof.* Take  $m = 2$  and  $B = c/2$  in Theorem 3.1. At  $T = 0$  the linearized coefficient in  $V - 1$  is  $2 - \eta$ . Comparing the coefficients of  $T$  and  $T^2$  gives (3.10)–(3.11). At every higher order the new unknown appears linearly with that same coefficient; the remaining expression is polynomial in previous coefficients and finitely many  $r_k$ , with polynomial dependence on  $\eta$ . Induction proves the rational-function assertion.  $\square$

**Proposition 3.3** (Geometric chart truncation). *Choose  $r, s, C > 0$  so that  $V$  is holomorphic on a neighborhood of the closed bidisk  $|T| \leq r$ ,  $|\eta| \leq s$  and  $|V| \leq C$  there. Write  $V = \sum_{j,k \geq 0} v_{jk}T^j\eta^k$ . If  $q_T = |T|/r < 1$  and  $q_\eta = |\eta|/s < 1$ , then*

$$\left| x - T \sum_{j=0}^J \sum_{k=0}^K v_{jk}T^j\eta^k \right| \leq |T|C \frac{q_T^{J+1} + q_\eta^{K+1}}{(1-q_T)(1-q_\eta)}. \quad (3.12)$$

*Proof.* Cauchy's inequalities give  $|v_{jk}| \leq Cr^{-j}s^{-k}$ . Sum the two geometric tails for  $j > J$  and for  $k > K$ ; counting their intersection twice only increases the bound.  $\square$

The theorem provides the existence of such a bidisk. Obtaining numerical values of  $r, s, C$  from a chosen bound on  $R$  is a separate local majorant calculation; the supplied numerical program does not certify those values.

### 3.3 What the chart says about ramification and radius

**Proposition 3.4** (Logarithmic obstruction to finite Puiseux ramification). *At criticality,*

$$x \sim 2\sqrt{\frac{\varepsilon}{c \log(1/\varepsilon)}}. \quad (3.13)$$

*There is no positive integer  $N$  for which the small critical inverse  $x(\varepsilon)$  is represented by a nonzero convergent power series in  $\varepsilon^{1/N}$  near 0. The radius of convergence of  $U_{\beta_c}(q)$  at  $q = 0$  is  $\rho_c$ .*

*Proof.* The defining equation  $v - \log v = \log(ce^{2h}/(4\varepsilon))$  gives  $v \sim \log(1/\varepsilon)$ . Since  $V \rightarrow 1$ , (3.13) follows. A convergent Puiseux series vanishing at zero has a first nonzero term  $C\varepsilon^{k/N}$ . The logarithmic order in (3.13) forces  $k/N = 1/2$ , but  $x/\sqrt{\varepsilon} \rightarrow 0$ , contradicting  $C \neq 0$ .

For  $0 < r < 1$ , positivity and (2.5) give  $u_n(\beta_c) \leq e^{n\beta_c A(r)}/(nr^n)$ . Letting  $r \uparrow 1$  shows that the radius is at least  $\rho_c$ . On  $0 < t < 1$ ,

$$\frac{d}{dt}(te^{-\beta_c A(t)}) = e^{-\beta_c A(t)}(1 - \beta_c t A'(t)) > 0,$$

because  $tA'(t) < d_1$ . The inverse branch reaches  $t = 1$  when  $q = \rho_c$ . If  $U_{\beta_c}$  were holomorphic there, then  $x = \beta_c d_0 + \varepsilon - U_{\beta_c}(q)$  would be holomorphic in  $\varepsilon$  at 0, contrary to the first part. Thus the radius is exactly  $\rho_c$ .  $\square$

The obstruction is logarithmic, not an analytic critical point with a new integer multiplicity. The critical inverse is nonetheless described by an *actually convergent* analytic function of two explicit small Lambert coordinates. Expanding  $W_{-1}$  itself produces the familiar nested logarithms; that optional further expansion is not needed for (3.12).

## 4 Lambert normalization of the sector scale

Let  $\beta$  range over a fixed compact subinterval of  $(0, \infty)$ . For sufficiently large  $n$ , define the large positive solution of

$$b_n^2 = nc_\beta(\log b_n + h), \quad \ell_n = \log b_n + h, \quad e_n = \sqrt{nc_\beta}. \quad (4.1)$$

The letter  $e_n$  denotes the *extreme-action scale*, not Euler's constant or an exponential function. These quantities have the exact form

$$\ell_n = -\frac{1}{2}W_{-1}\left(-\frac{2e^{-2h}}{nc_\beta}\right), \quad b_n = e_n\sqrt{\ell_n}. \quad (4.2)$$

Indeed,  $e^{2\ell_n - 2h} = nc_\beta \ell_n$  is equivalent to the Lambert equation. In particular,

$$\ell_n \sim \frac{1}{2} \log n, \quad b_n \sim \sqrt{\frac{nc_\beta \log n}{2}}, \quad \frac{e_n}{b_n} = \ell_n^{-1/2} \longrightarrow 0. \quad (4.3)$$

The constants  $h$  and  $c_\beta$  are retained in (4.2); replacing them prematurely by a leading logarithm changes the correction coefficients below.

For a general integer  $k$ , define

$$\lambda = \frac{k - n\beta d_1}{b_n}. \quad (4.4)$$

For the desired sector coefficient,  $k = n$  and  $\lambda_n = n(1 - \beta d_1)/b_n$ . Thus a compact critical window  $|\lambda_n| \leq R$  has coupling width

$$\beta - \beta_c = O\left(\sqrt{\frac{\log n}{n}}\right). \quad (4.5)$$

We shall allow  $\beta = \beta_n$  to vary within this window. The all-order local-limit theorem itself is stronger: it is uniformly additive for every integer  $k$  and for  $\beta$  in a fixed compact positive interval.

## 5 An all-order Gaussian–logarithmic sector expansion

### 5.1 The correction kernels

Let  $\phi(\lambda) = (2\pi)^{-1/2}e^{-\lambda^2/2}$  and define, for  $j \geq 0$ ,

$$G_j(\lambda) = \frac{1}{2\pi 2^j j!} \int_{\mathbb{R}} e^{-s^2/2 - i\lambda s} s^{2j} [\text{Log}(-is)]^j ds. \quad (5.1)$$

At  $s = 0$  the integrand is interpreted by its integrable limit when  $j \geq 1$ . The branch is  $\text{Log}(-is) = \log|s| - i(\pi/2) \text{sgn}(s)$ . The integrals define entire functions of  $\lambda$ : on each compact complex set, the Gaussian dominates every derivative of the integrand. They are real on the real axis by conjugation symmetry, and  $G_0 = \phi$ .

**Theorem 5.1** (Uniform all-order local limit). *For each integer  $K \geq 0$ , uniformly for  $\beta$  in a compact subinterval of  $(0, \infty)$  and all integers  $k$ ,*

$$b_n \mathbb{P}(S_{n,\beta} = k) = \sum_{j=0}^K \frac{G_j(\lambda)}{\ell_n^j} + O(\ell_n^{-K-1}), \quad \lambda = \frac{k - n\beta d_1}{b_n}. \quad (5.2)$$

Consequently, for  $k = n$ ,

$$u_n(\beta) = \frac{\rho_\beta^{-n}}{nb_n} \left\{ \sum_{j=0}^K \frac{G_j(\lambda_n)}{\ell_n^j} + O(\ell_n^{-K-1}) \right\}. \quad (5.3)$$

These estimates are additive. Dividing by  $\phi(\lambda)$  gives a uniform relative expansion only on compact real  $\lambda$  intervals.

### 5.2 Fourier damping and a complete proof

**Lemma 5.2** (Aperiodic logarithmic damping). *There are positive constants  $C_1, C_2, \theta_0$  such that*

$$\sum_{j \geq 1} a_j (1 - \cos j\theta) \geq C_1 \theta^2 \log(1/|\theta|) \quad (0 < |\theta| \leq \theta_0), \quad (5.4)$$

$$\sum_{j \geq 1} a_j (1 - \cos j\theta) \geq C_2 \quad (\theta_0 \leq |\theta| \leq \pi). \quad (5.5)$$

For the truncated sum, the first lower bound is replaced, after reducing  $\theta_0$  and taking  $M$  large, by

$$\sum_{j \leq M} a_j (1 - \cos j\theta) \geq C_1 \theta^2 \log(\min(M, 1/|\theta|)). \quad (5.6)$$

The away-from-zero estimate remains valid for  $M \geq 1$ .

*Proof.* For  $|j\theta| \leq 1$ ,  $1 - \cos(j\theta) \geq c_0 j^2 \theta^2$ . Sum over the unaltered tail indices  $J < j \leq \min(M, 1/|\theta|)$  and use  $\sum_{J < j \leq L} j^{-1} \geq \frac{1}{2} \log L$  for large  $L$ . This proves the small-angle bounds with fixed smaller constants. For  $\theta_0 \leq |\theta| \leq \pi$ , the term  $a_1(1 - \cos \theta)$  has a strictly positive minimum.  $\square$

*Proof of Theorem 5.1.* Fourier inversion gives

$$b_n \mathbb{P}(S = k) = \frac{1}{2\pi} \int_{-\pi b_n}^{\pi b_n} e^{-i\lambda s} \exp\{\Psi_n(s)\} ds, \quad (5.7)$$

where

$$\Psi_n(s) = n\beta \left( A(e^{is/b_n}) - d_0 - id_1 s/b_n \right).$$

From (3.3), with  $x = -is/b_n$ , and the defining equality  $b_n^2 = nc_\beta \ell_n$ , we obtain

$$\Psi_n(s) = -\frac{s^2}{2} + \frac{s^2}{2\ell_n} \operatorname{Log}(-is) + O\left(\frac{n|s|^3}{b_n^3}\right) \quad (5.8)$$

uniformly when  $|s|/b_n$  is sufficiently small. The analytic remainder in (3.3) is uniformly bounded as  $\beta$  varies in the specified compact interval.

We first justify localization. For  $|s| \leq b_n^{1/2}$ , the logarithm in (5.4) is at least  $(\log b_n)/2$ , so the modulus of  $e^{\Psi_n(s)}$  is bounded by  $e^{-Cs^2}$ . For  $b_n^{-1/2} \leq |\theta| \leq \theta_0$ , the function  $\theta^2 \log(1/|\theta|)$  is increasing after choosing  $\theta_0$  small. The damping exponent is at least a positive multiple of  $b_n$ . For  $\theta_0 \leq |\theta| \leq \pi$  it is at least a positive multiple of  $n$ . Their contributions to (5.7), including the factor  $b_n$ , are smaller than every power of  $1/\ell_n$ .

Fix  $K$  and choose  $D$  large. On  $|s| \leq S_n = \sqrt{D \log \ell_n}$ , the last term of (5.8) is negligible to every fixed inverse-logarithmic order, since  $n/b_n^3 = O(n^{-1/2}(\log n)^{-3/2})$ . Expand

$$\exp\left(\frac{s^2}{2\ell_n} \operatorname{Log}(-is)\right) = \sum_{j=0}^K \frac{s^{2j} [\operatorname{Log}(-is)]^j}{2^j j! \ell_n^j} + \mathcal{E}_{K,n}(s).$$

The integral of  $e^{-s^2/2} |\mathcal{E}_{K,n}(s)|$  on this interval is  $O(\ell_n^{-K-1})$ . To see this without a growing logarithmic loss, use the exponential Taylor remainder. Its extra factor

$$\exp(O(s^2 |\operatorname{Log}(-is)|/\ell_n))$$

can be absorbed into  $e^{-s^2/4}$ ; the resulting integral of  $|s|^{2K+2}(1 + |\log |s||)^{K+1} e^{-s^2/4}$  is finite and independent of  $n$ . The analytic remainder in (5.8) is handled in the same way.

The part  $S_n < |s| \leq b_n^{1/2}$  is  $O(\ell_n^{-K-1})$  by the Gaussian damping, after increasing  $D$ . The tails of the individual kernel integrals satisfy the same estimate. This gives (5.2). Every bound is independent of  $\lambda$ , because  $|e^{-i\lambda s}| = 1$  for real  $\lambda$ . Finally use (2.6) to obtain (5.3).  $\square$

### 5.3 Explicit critical coefficients

**Corollary 5.3** (First critical corrections and a formula at every order). *At  $\beta = \beta_c$ ,*

$$u_n(\beta_c) = \frac{\rho_c^{-n}}{\sqrt{2\pi} n b_n} \left( 1 + \frac{g_1}{\ell_n} + \frac{g_2}{\ell_n^2} + \cdots + \frac{g_K}{\ell_n^K} + O(\ell_n^{-K-1}) \right), \quad (5.9)$$

where, with  $\gamma$  denoting Euler's constant,

$$g_1 = \frac{2 - \gamma - \log 2}{4}, \quad (5.10)$$

$$g_2 = \frac{3}{32} \left[ \left( \frac{8}{3} - \gamma - \log 2 \right)^2 - \frac{\pi^2}{2} - \frac{40}{9} \right]. \quad (5.11)$$

For every  $j \geq 0$ , define

$$M_j(z) = \frac{2^{j+z/2} \Gamma(j + (z+1)/2)}{\sqrt{\pi}}.$$

Then

$$g_j = \frac{1}{2^j j!} \Re \left[ \left( \frac{d}{dz} - \frac{i\pi}{2} \right)^j M_j(z) \right]_{z=0}. \quad (5.12)$$

In particular,

$$u_n(\beta_c) \sim \frac{\rho_c^{-n}}{\sqrt{\pi c} n^{3/2} \sqrt{\log n}}. \quad (5.13)$$

*Proof.* At criticality  $\lambda_n = 0$ . Divide (5.1) by  $\phi(0)$  and use the even symmetry to write it as  $(2^j j!)^{-1} \mathbb{E}[|Z|^{2j} \Re(\log |Z| - i\pi/2)^j]$ , where  $Z$  is standard normal. Its Mellin moment is  $M_j(z)$ ; differentiation under the integral is justified by Gaussian decay and integrability at zero. This proves (5.12). The elementary half-integer gamma recurrences yield (5.10) and (5.11). Finally use (4.3).  $\square$

The finite prefix changes  $d_0, d_1, h$ , and therefore changes the exponential factor and exact Lambert scale. After that normalization, the kernels  $G_j$  do not depend on the prefix. Its remaining analytic information, carried by  $R_\beta$ , contributes terms of order  $n/b_n^3$  and smaller in the Fourier exponent. These are beyond all fixed inverse-logarithmic orders, but they need not be absent. This is a precise boundary of the universality statement.

## 6 Sharp action budgets and finite-coupling cutoff corrections

Throughout this section,  $\beta = \beta_n$  lies in a fixed compact positive interval, and the sector coordinate  $\lambda_n = n(1 - \beta_n d_1)/b_n$  lies in a fixed compact real interval. In particular,  $\beta_n \rightarrow \beta_c$ . Use the scales in (4.1) with this varying  $\beta_n$ , and put

$$s_n = \frac{M}{e_n}. \quad (6.1)$$

The first theorem gives the leading law. Its refinement then quantifies why convergence to that law is slow.

**Theorem 6.1** (Cutoff profile and necessary-and-sufficient budget). *Uniformly when  $s_n$  and  $\lambda_n$  range over compact subsets of  $(0, \infty)$  and  $\mathbb{R}$ , respectively,*

$$R_{n,M}(\beta_n) = \exp\left(-\frac{1}{2s_n^2}\right) + o(1). \quad (6.2)$$

For arbitrary integer cutoffs  $M = M_n \geq 1$ , subject to the same compact critical-window condition,

$$\boxed{R_{n,M_n}(\beta_n) \rightarrow 1 \iff M_n/e_n \rightarrow \infty.} \quad (6.3)$$

If  $M_n/e_n \rightarrow 0$ , the retained fraction tends to zero. Thus the sharp budget scale is  $e_n \asymp \sqrt{n}$ , not  $b_n \asymp \sqrt{n \log n}$ .

### 6.1 A quantitative local limit for the retained actions

The exact omitted mass, omitted mean on the full fluctuation scale, and retained variance on that scale are

$$\nu_{n,M} = n\beta Q_M, \quad \kappa_{n,M} = \frac{n\beta}{b_n} \sum_{j>M} j a_j, \quad v_{n,M} = \frac{n\beta}{b_n^2} \sum_{j\leq M} j^2 a_j. \quad (6.4)$$

They must not be confused with the exponentially tilted variance of a finite fold in Section 8. Let  $\phi_v(z) = (2\pi v)^{-1/2} \exp(-z^2/(2v))$ .

**Lemma 6.2** (Truncated Gaussian approximation with a rate). *If  $s_n = M/e_n$  ranges over a compact subset of  $(0, \infty)$ , then, uniformly for all integers  $k$ ,*

$$b_n \mathbb{P}(S_M = k) = \phi_{v_{n,M}} \left( \frac{k - n\beta d_1}{b_n} + \kappa_{n,M} \right) + O(\ell_n^{-3/2}). \quad (6.5)$$

*Proof.* Write  $\mu_M = n\beta \sum_{j \leq M} j a_j$ . The centered characteristic exponent, at frequency  $s/b_n$ , equals

$$n\beta \sum_{j \leq M} a_j (e^{ijs/b_n} - 1 - ijs/b_n) = -\frac{v_{n,M} s^2}{2} + \mathcal{R}_n(s),$$

with

$$|\mathcal{R}_n(s)| \leq \frac{n\beta |s|^3}{6b_n^3} \sum_{j \leq M} j^3 a_j = O(\ell_n^{-3/2} |s|^3). \quad (6.6)$$

Here  $\sum_{j \leq M} j^3 a_j = aM + O(1)$  and  $M \asymp e_n$ . Also

$$v_{n,M} = \frac{H_M + p^{[2]}/a}{\ell_n} \longrightarrow 1, \quad H_M = \sum_{j=1}^M \frac{1}{j},$$

because  $\log M = \log b_n - \frac{1}{2} \log \ell_n + O(1)$ .

For  $|s| \leq \sqrt{D \log \ell_n}$ , integrate the difference between the true centered characteristic function and  $e^{-v_{n,M} s^2/2}$ . Using (6.6), the Gaussian damping, and the identity  $e^z - e^w = (z - w) \int_0^1 e^{(1-t)w + tz} dt$ , its integral is  $O(\ell_n^{-3/2})$ . The remainder of the Fourier interval is negligible to this order. Indeed, for  $|\theta| \leq M^{-1/2}$ , (5.6) gives a Gaussian bound on the  $b_n$  scale, since  $\log \min(M, 1/|\theta|) \geq \frac{1}{2} \log M \asymp \ell_n$ . For  $M^{-1/2} \leq |\theta| \leq \theta_0$  the exponent is at least a constant times  $n \log M/M$ , which is much larger than  $\log b_n$ . The remaining interval is controlled by  $a_1 > 0$ . Fourier inversion now gives (6.5); the shift is  $(k - \mu_M)/b_n = (k - n\beta d_1)/b_n + \kappa_{n,M}$ . The absolute integral bound is independent of  $k$ .  $\square$

**Theorem 6.3** (Uniform refined cutoff formula). *Put  $g(\lambda) = G_1(\lambda)/\phi(\lambda)$ . Uniformly for  $s_n$  in a compact positive interval and  $\lambda_n$  in a compact real interval,*

$$\boxed{\log R_{n,M} = -\nu_{n,M} - \frac{1}{2} \log v_{n,M} + \frac{\lambda_n^2}{2} - \frac{(\lambda_n + \kappa_{n,M})^2}{2v_{n,M}} - \frac{g(\lambda_n)}{\ell_n} + O(\ell_n^{-3/2})}. \quad (6.7)$$

For  $M > J$ , define

$$D_n = \frac{1}{2} \log \ell_n - \log s_n - \gamma + \frac{3}{2}. \quad (6.8)$$

Then the equivalent expanded form is

$$\boxed{\log R_{n,M} = -\frac{1}{2s_n^2} - \frac{\lambda_n}{s_n \sqrt{\ell_n}} + \frac{1}{\ell_n} \left( \frac{1 - \lambda_n^2}{2} D_n - \frac{1}{2s_n^2} - g(\lambda_n) \right) + O\left(\frac{\log \ell_n}{\ell_n^{3/2}}\right)}. \quad (6.9)$$

*Proof.* Theorem 5.1 with  $K = 1$  gives

$$b_n \mathbb{P}(S = n) = \phi(\lambda_n) \left( 1 + \frac{g(\lambda_n)}{\ell_n} + O(\ell_n^{-2}) \right).$$

Lemma 6.2 gives the corresponding numerator. On the specified compact sets the Gaussian values stay bounded away from zero, because  $v_{n,M} \rightarrow 1$  and  $\kappa_{n,M} \rightarrow 0$ . Taking logarithms in (2.7) proves (6.7).

For  $M > J$ , integral comparison or Euler summation gives

$$\begin{aligned}\sum_{j>M} a_j &= a \left( \frac{1}{2M^2} + O(M^{-3}) \right), \\ \sum_{j>M} j a_j &= a \left( \frac{1}{M} + O(M^{-2}) \right), \\ \sum_{j \leq M} j^2 a_j &= a(\log M + \gamma) + p^{[2]} + O(M^{-1}).\end{aligned}$$

Using  $M = s_n e_n$ ,  $b_n = e_n \sqrt{\ell_n}$ , and  $\ell_n = \log b_n + 3/2 + p^{[2]}/a$  yields

$$\nu_{n,M} = \frac{1}{2s_n^2} + O(n^{-1/2}), \quad (6.10)$$

$$\kappa_{n,M} = \frac{1}{s_n \sqrt{\ell_n}} + O(n^{-1/2} \ell_n^{-1/2}), \quad (6.11)$$

$$v_{n,M} = 1 - \frac{D_n}{\ell_n} + O(n^{-1/2} \ell_n^{-1}). \quad (6.12)$$

In particular the explicit  $p^{[2]}$  cancels in  $D_n$ . Expand the logarithm and reciprocal of  $v_{n,M}$  in (6.7). The first cross term from  $\kappa_{n,M}$  is  $-\lambda_n/(s_n \sqrt{\ell_n})$ ; the square of that shift contributes  $-1/(2s_n^2 \ell_n)$ . The linear variance correction is  $(1 - \lambda_n^2)D_n/(2\ell_n)$ . The unused products are bounded by  $O(D_n \ell_n^{-3/2} + D_n^2 \ell_n^{-2})$ , which is  $O((\log \ell_n) \ell_n^{-3/2})$  for large  $n$ . This proves (6.9).  $\square$

At exact criticality,  $\lambda_n = 0$ , so (6.9) simplifies to

$$\log R_{n,M} = -\frac{1}{2s_n^2} + \frac{1}{\ell_n} \left[ \frac{1}{4} \log \ell_n - \frac{1}{2} \log s_n + \frac{1 - \gamma + \log 2}{4} - \frac{1}{2s_n^2} \right] + O\left(\frac{\log \ell_n}{\ell_n^{3/2}}\right). \quad (6.13)$$

The  $(\log \ell_n)/\ell_n$  term explains a particularly slow approach to the leading cutoff profile. Away from  $\lambda_n = 0$ , the larger  $1/\sqrt{\ell_n}$  correction records the omitted mean. It cannot be recovered from an omitted-intensity estimate alone.

*Proof of Theorem 6.1.* The leading formula follows immediately from Theorem 6.3. To extend it to arbitrary cutoffs, use monotonicity in (2.7). If  $M_n/e_n \rightarrow \infty$ , then for every fixed  $L > 0$ ,

$$\liminf_{n \rightarrow \infty} R_{n,M_n} \geq \lim_{n \rightarrow \infty} R_{n, \lfloor L e_n \rfloor} = e^{-1/(2L^2)}.$$

Let  $L \rightarrow \infty$  and use  $R_{n,M_n} \leq 1$ . Conversely, if  $M_n/e_n$  does not tend to infinity, some subsequence has  $M_n/e_n \leq L$  for a finite  $L$ . Along it, monotonicity bounds the limsup by  $e^{-1/(2L^2)} < 1$  (enlarging  $L$  slightly if necessary). Finally, if  $M_n/e_n \rightarrow 0$ , bound above by the profile at  $\lfloor L e_n \rfloor$  for every fixed  $L > 0$  and then let  $L \downarrow 0$ .  $\square$

## 6.2 Action selection and the limit of the guarantee

For a desired asymptotic retained fraction  $r \in (0, 1)$ , a natural budget is

$$M_n \sim \frac{\sqrt{n \beta_n a}}{\sqrt{2 \log(1/r)}}. \quad (6.14)$$

For example, as the allowed loss  $1 - r$  tends to zero after the asymptotic limit, this factor behaves like  $1/\sqrt{2(1-r)}$ . Formula (6.7) gives a more informative finite- $n$  prediction, because its omitted sums can be evaluated exactly or tightly bounded. It is not, as presently implemented, a certified finite- $n$  relative-error algorithm: explicit constants and threshold values in the  $O$  term would be required for that stronger deliverable.

## 7 The conditional point process of large actions

The cutoff profile has a transparent interpretation. Although a large individual action has size of order  $e_n$ , its contribution divided by  $b_n$  tends to zero. Conditioning on a fluctuation in the  $b_n$  window therefore leaves the leading extreme-action process unchanged. This is the action-count analogue of the classical conditioning/extremes phenomenon discussed in [4].

**Theorem 7.1** (Poisson limit for conditioned large actions). *Under the compact critical-window assumptions of Section 6, condition on  $S_{n,\beta_n} = n$ . The point process*

$$\Xi_n = \sum_{j \geq 1} Z_j \delta_{j/e_n} \quad (7.1)$$

*converges vaguely on  $(0, \infty)$  to a Poisson point process with intensity  $x^{-3} dx$ . More explicitly, for every nonnegative continuous function  $f$  with compact support in  $(0, \infty)$ ,*

$$\mathbb{E} \left[ e^{-\int f d\Xi_n} \mid S = n \right] \longrightarrow \exp \left( \int_0^\infty (e^{-f(x)} - 1) x^{-3} dx \right), \quad (7.2)$$

*uniformly for  $\lambda_n$  in a fixed compact interval. For every  $s > 0$ , the conditioned number of actions larger than  $se_n$  converges to  $\text{Pois}(1/(2s^2))$ .*

*Proof.* The joint Fourier transform with the Laplace weight differs from the unweighted transform by the factor

$$\exp \left( n\beta_n \sum_{j \geq 1} a_j (e^{-f(j/e_n)} - 1) e^{ijs/b_n} \right). \quad (7.3)$$

For  $s$  in a bounded interval,  $j/e_n$  remains in a fixed compact subset of  $(0, \infty)$  wherever  $f$  is nonzero. Since  $e_n/b_n \rightarrow 0$ ,  $e^{ijs/b_n} \rightarrow 1$  uniformly there. The Riemann sum, using  $n\beta_n a = e_n^2$ , tends to  $\int (e^{-f(x)} - 1) x^{-3} dx$ .

The exponent of (7.3) has uniformly bounded modulus on the entire Fourier interval: the sum of its absolute coefficients is  $O(n\beta_n \sum_{j \geq ce_n} a_j) = O(1)$ , where  $c > 0$  is below the support of  $f$ . Thus the unweighted Fourier damping is preserved up to a constant factor. Dominated localization as in Theorem 5.1 gives the limiting Gaussian integral times the limiting factor in (7.3). Division by  $\mathbb{P}(S = n)$  proves (7.2), with uniformity on compact  $\lambda_n$  intervals.

To pass from compactly supported tests to counts on  $(s, \infty)$ , control the far tail. The Poisson shift identity gives

$$\mathbb{E} \left[ \sum_{j > Re_n} Z_j \mid S = n \right] = n\beta_n \sum_{j > Re_n} a_j \frac{\mathbb{P}(S = n - j)}{\mathbb{P}(S = n)}.$$

The uniformly additive local limit bounds the numerator by  $C/b_n$  for every  $j$ , while the denominator is at least  $c_0/b_n$  in the compact window. Hence the displayed expectation is  $O(R^{-2})$ , uniformly in large  $n$ . This proves tightness of the far tail and the asserted Poisson count limit.  $\square$

If  $J_{(k)}$  is the  $k$ th largest action, counting multiplicity and setting it to zero when fewer than  $k$  actions occur, then

$$\mathbb{P}(J_{(k)} \leq se_n \mid S = n) \longrightarrow e^{-\mu_s} \sum_{r=0}^{k-1} \frac{\mu_s^r}{r!}, \quad \mu_s = \frac{1}{2s^2}. \quad (7.4)$$

The case  $k = 1$  reproduces (6.2). This derivation also shows which step fails in the fractional stable regime: there the extreme and fluctuation scales are comparable, so the Fourier phase of a large action does not disappear.

## 8 Moving finite-action folds: a convergent all-order drift chart

At  $\beta = \beta_c$ , every finite truncation has an ordinary analytic fold, even though the limiting critical point is logarithmic. The location of that fold determines the exponentially small coefficient loss when the cutoff lies below the extreme scale. It is essential to distinguish its exact location from a truncated asymptotic approximation.

### 8.1 Critical point, critical value, and curvature

For  $M > J$ , let  $\tau_M > 1$  be the unique solution of

$$\beta_c \sum_{j \leq M} j a_j \tau_M^j = 1. \quad (8.1)$$

Existence and uniqueness follow because the left side is strictly increasing, is below 1 at  $\tau = 1$ , and tends to infinity. Define

$$\sigma_M = \log \tau_M, \quad y_M = M \sigma_M, \quad r_M = \tau_M e^{-\beta_c A_M(\tau_M)}, \quad \mathcal{V}_M = \beta_c \sum_{j \leq M} j^2 a_j \tau_M^j, \quad (8.2)$$

and

$$L_M = H_M + \frac{p^{[2]}}{a}. \quad (8.3)$$

The second derivative of  $\log q$  with respect to  $\log t$  at the critical point is  $-\mathcal{V}_M < 0$ , so this is a nondegenerate analytic fold. The use of  $\mathcal{V}_M$  here avoids confusion with the untilted quantity  $v_{n,M}$  in (6.4).

Introduce entire functions

$$F(y) = \sum_{k \geq 2} \frac{y^k}{k!(k-1)} = \int_0^1 \frac{e^{yt} - 1 - yt}{t^2} dt, \quad (8.4)$$

$$G(y) = \sum_{k \geq 3} \frac{y^k}{k!(k-2)}, \quad G'(y) = F(y), \quad (8.5)$$

$$J(y) = \sum_{k \geq 1} \frac{y^k}{k!k}. \quad (8.6)$$

The removable singularities at  $t = 0$  in the integral representation are understood by Taylor expansion.

**Theorem 8.1** (Convergent finite-fold drift functions). *There is a unique function  $Y(z)$  analytic near  $z = 0$  such that*

$$Y = z(1 - F(Y)), \quad Y(0) = 0. \quad (8.7)$$

Define the analytic function, with its removable value at  $z = 0$ ,

$$\mathcal{H}(z) = \frac{1}{2} + Y(z) - \frac{Y(z)^2}{2z} - G(Y(z)). \quad (8.8)$$

Then

$$y_M = Y(1/L_M) + O((ML_M)^{-1}), \quad (8.9)$$

$$\frac{M^2}{c} \log \frac{r_M}{\rho_c} = \mathcal{H}(1/L_M) + O(M^{-1}), \quad (8.10)$$

$$\frac{\mathcal{V}_M}{c} = L_M + J(Y(1/L_M)) + O(M^{-1}). \quad (8.11)$$

Thus the complete inverse-logarithmic expansions of these quantities are obtained by substituting  $z = 1/L_M$  into convergent power series.

*Proof.* The implicit equation  $Y - z(1 - F(Y)) = 0$  has derivative 1 in  $Y$  at  $(Y, z) = (0, 0)$ , so the analytic implicit function theorem proves the first assertion. Since  $Y(z) = z + O(z^3)$ , the quotient in (8.8) is removable and  $\mathcal{H}$  is analytic.

Subtract (8.1) at  $\tau = 1$  from its value at  $\tau_M$ . The tail is unaltered for  $M > J$ , so

$$\sum_{j \leq M} ja_j(e^{jy_M/M} - 1) = a \sum_{j > M} j^{-2}. \quad (8.12)$$

The linear term on the left is  $ay_M L_M/M$ . For bounded  $y \geq 0$ , the remaining tail terms, after multiplication by  $M/a$ , form a Riemann sum for  $F(y)$ , with error  $O(y^2/M)$ ; the finite-prefix nonlinear terms have the same error order. More explicitly, the function  $(e^{yt} - 1 - yt)/t^2$  extends smoothly to  $[0, 1]$ , and its supremum and first derivative there are  $O(y^2)$  for bounded  $y$ . Meanwhile  $M \sum_{j > M} j^{-2} = 1 + O(M^{-1})$ . It follows that

$$L_M y_M + F(y_M) = 1 + O(M^{-1}). \quad (8.13)$$

Positivity in (8.12) gives  $y_M = O(1/L_M)$  (the total linear moment is positive for large  $M$ ). The exact equation  $L_M y + F(y) = 1$  has the solution  $Y(1/L_M)$ . Its derivative on the intervening small positive interval is  $L_M + F'(y) \geq L_M/2$  for large  $M$ . The mean value theorem proves (8.9).

For the critical value, cancel the first-order terms before estimating:

$$\log \frac{r_M}{\rho_c} = \beta_c \sum_{j > M} a_j + \beta_c \sigma_M \sum_{j > M} ja_j - \beta_c \sum_{j \leq M} a_j(e^{j\sigma_M} - 1 - j\sigma_M). \quad (8.14)$$

Its quadratic term is  $-cy_M^2 L_M/(2M^2)$ . The terms of order at least three give a Riemann sum for  $-cG(y_M)/M^2$ , with error  $O(M^{-3})$  for the bounded  $y_M$  under consideration. Using  $M^2 \sum_{j > M} j^{-3} = 1/2 + O(M^{-1})$  and  $M \sum_{j > M} j^{-2} = 1 + O(M^{-1})$ , we obtain

$$\frac{M^2}{c} \log \frac{r_M}{\rho_c} = \frac{1}{2} + y_M - \frac{L_M y_M^2}{2} - G(y_M) + O(M^{-1}).$$

Replace  $y_M$  by  $Y(1/L_M)$  using (8.9). The replacement error is  $O(M^{-1})$ ; in fact the first derivative of the main expression vanishes at that exact solution. This proves (8.10).

Finally,

$$\frac{\mathcal{V}_M}{c} = L_M + \sum_{j \leq M} \frac{e^{jy_M/M} - 1}{j} + O(y_M/M).$$

The finite-prefix term has been separated, and the sum is a Riemann sum for  $J(y_M)$  with error  $O(M^{-1})$ . Substituting (8.9) proves (8.11).  $\square$

## 8.2 Coefficients, cancellation, and explicit drift laws

The first terms of the convergent functions are

$$Y(z) = z - \frac{1}{2}z^3 - \frac{1}{12}z^4 + \frac{35}{72}z^5 + \frac{33}{160}z^6 - \frac{1013}{1800}z^7 + O(z^8), \quad (8.15)$$

$$\mathcal{H}(z) = \frac{1}{2} + \frac{z}{2} - \frac{z^3}{6} - \frac{z^4}{48} + \frac{11z^5}{90} + \frac{119z^6}{2880} + O(z^7), \quad (8.16)$$

$$J(Y(z)) = z + \frac{z^2}{4} - \frac{4z^3}{9} - \frac{31z^4}{96} + \frac{653z^5}{1800} + O(z^6). \quad (8.17)$$

They follow by substitution in (8.7) and ordinary power-series arithmetic. In particular,

$$ML_M \log \tau_M = 1 - \frac{1}{2L_M^2} - \frac{1}{12L_M^3} + O(L_M^{-4} + M^{-1}), \quad (8.18)$$

$$\log \frac{r_M}{\rho_c} = \frac{c}{2M^2} \left( 1 + \frac{1}{L_M} - \frac{1}{3L_M^3} + O(L_M^{-4} + M^{-1}) \right), \quad (8.19)$$

$$\mathcal{V}_M = c \left( L_M + \frac{1}{L_M} + \frac{1}{4L_M^2} + O(L_M^{-3} + M^{-1}) \right). \quad (8.20)$$

There is no  $L_M^{-2}$  term in the bracket in (8.19). That cancellation is not visible from the leading estimate  $\sigma_M \asymp (M \log M)^{-1}$  alone. The single analytic function  $\mathcal{H}$  determines every later cancellation and correction.

*Remark 8.2* (Why convergence here does not imply convergence everywhere). The series for  $Y$ ,  $\mathcal{H}$ , and  $J \circ Y$  converge near 0. After substituting  $1/L_M$ , they describe the logarithmic part of the finite- $M$  expansion, but an additional error of order  $M^{-1}$  remains in the normalized formulas. This is an analytic two-scale expansion, not an assertion that a power series in  $1/\log M$  represents the finite fold exactly for each  $M$ .

## 8.3 A uniform saddle law for finite truncations

**Theorem 8.3** (Triangular finite-fold coefficient law). *At  $\beta = \beta_c$ , suppose  $M = M_n \rightarrow \infty$  and*

$$\frac{M_n}{\sqrt{n \log M_n}} \rightarrow 0. \quad (8.21)$$

*With the exact quantities in (8.2),*

$$u_n^{(M)}(\beta_c) \sim \frac{r_M^{-n}}{\sqrt{2\pi n^{3/2} \mathcal{V}_M}}. \quad (8.22)$$

*Proof.* Exponentially tilt the finite Poisson family to independent variables with means  $n\beta_c a_j \tau_M^j$ ,  $j \leq M$ . Its sum  $\tilde{S}_M$  has mean  $n$  by (8.1) and variance  $n\mathcal{V}_M$ . Lagrange inversion and exponential tilting give exactly

$$u_n^{(M)}(\beta_c) = \frac{r_M^{-n}}{n} \mathbb{P}(\tilde{S}_M = n). \quad (8.23)$$

Set  $B_n = \sqrt{n\mathcal{V}_M} \asymp \sqrt{n \log M}$ . Since  $\tau_M^j \leq e^{y_M} = 1 + o(1)$ , the third absolute cumulant divided by  $B_n^3$  is at most a constant times

$$\frac{nM}{(n \log M)^{3/2}} = O\left(\frac{M/B_n}{\log M}\right) \rightarrow 0.$$

For every fixed Fourier frequency on the  $B_n$  scale, the centered characteristic function therefore tends to  $e^{-s^2/2}$ .

For completeness, the local limit also requires tail control. The tilt only increases the positive weights, so Lemma 5.2 applies. On  $|\theta| \leq M^{-1/2}$ , its truncated lower bound is at least a constant times  $\theta^2 \log M$ , giving  $e^{-Cs^2}$  on the  $B_n$  scale. On  $M^{-1/2} \leq |\theta| \leq \theta_0$ , it is at least a constant times  $n \log M/M$ . Under (8.21), this quantity dominates  $\log B_n$ . The remaining interval has damping of order  $n$ . Thus dominated Fourier inversion gives  $\mathbb{P}(\tilde{S}_M = n) \sim (2\pi n \mathcal{V}_M)^{-1/2}$ . Substitute in (8.23).  $\square$

The sufficient condition (8.21) includes  $M \asymp \sqrt{n}$ , where the finite-fold description overlaps with the cutoff profile. Dividing (8.22) by the full critical leading coefficient gives

$$R_{n,M} \sim \exp\left(-n \log \frac{r_M}{\rho_c}\right) \sqrt{\frac{c\ell_n}{\mathcal{V}_M}}. \quad (8.24)$$

At  $M = s_n e_n$  with  $s_n \rightarrow s > 0$ , (8.19) yields  $n \log(r_M/\rho_c) \rightarrow 1/(2s^2)$  and the square-root factor tends to 1. Hence the analytic moving-fold calculation agrees with the probabilistic cutoff law. The refined formula in Theorem 6.3 provides more precision than the leading saddle equivalence (8.24).

**Proposition 8.4** (When a drift approximation may be exponentiated). *Let  $\mathcal{H}_K$  be the Taylor polynomial of  $\mathcal{H}$  through degree  $K$ , and set*

$$\Delta_{M,K} = \frac{c}{M^2} \mathcal{H}_K(1/L_M).$$

Then

$$\log(r_M/\rho_c) - \Delta_{M,K} = O\left(M^{-2} L_M^{-K-1} + M^{-3}\right). \quad (8.25)$$

Replacing  $(r_M/\rho_c)^{-n}$  by  $e^{-n\Delta_{M,K}}$  incurs a relative error tending to zero whenever

$$\frac{n}{M^2 L_M^{K+1}} + \frac{n}{M^3} \rightarrow 0. \quad (8.26)$$

*Proof.* The analyticity of  $\mathcal{H}$  gives a Taylor remainder  $O(L_M^{-K-1})$ , and (8.10) gives the  $O(M^{-3})$  error after rescaling. Multiply by  $n$  before exponentiating.  $\square$

If only the leading drift  $c/(2M^2)$  is used, its first omitted term is  $c/(2M^2 L_M)$ , which may become large after multiplication by  $n$  in a hard-cutoff regime. A correct leading statement about  $r_M$  is not automatically a correct relative statement about  $r_M^{-n}$ .

## 9 A slowly varying extension: logarithmically weighted cubic tails

The exact polylogarithm is responsible for the clean all-order analytic remainder in (3.3). The leading scale separation needs much less. This section treats a concrete slowly varying class, without asserting the same all-order corrections under weaker hypotheses.

Assume now that  $a_j \geq 0$ ,  $a_1 > 0$ , and

$$a_j \sim a j^{-3} (\log j)^r \quad (j \rightarrow \infty), \quad a > 0, \quad r \geq -1. \quad (9.1)$$

Only large  $j$  are involved in this assumption. Define  $A(t) = \sum a_j t^j$ ,  $d_0 = A(1)$ ,  $d_1 = \sum j a_j$ ,  $\beta_c = d_1^{-1}$ , and  $\rho_\beta = e^{-\beta d_0}$  as before. The parametrization, Lagrange identity, and Poisson

formulas of Section 2 remain exact. Put  $L_n^* = \frac{1}{2} \log n$ , and use the following *leading-order* scales for  $\beta$  in a compact positive interval:

$$e_n^* = \sqrt{n\beta a} (L_n^*)^{r/2}, \quad (9.2)$$

$$b_n^* = \begin{cases} \sqrt{\frac{n\beta a}{r+1}} (L_n^*)^{(r+1)/2}, & r > -1, \\ \sqrt{n\beta a \log \log n}, & r = -1. \end{cases} \quad (9.3)$$

These are not exact Lambert scales and should not be substituted into the correction formulas of Sections 5–6.

**Lemma 9.1** (Tail and truncated-variance estimates). *Under (9.1), as  $x \rightarrow \infty$ ,*

$$\sum_{j>x} a_j \sim \frac{a}{2} x^{-2} (\log x)^r, \quad \sum_{j>x} j a_j \sim a x^{-1} (\log x)^r, \quad (9.4)$$

$$V(x) := \sum_{j \leq x} j^2 a_j \sim \begin{cases} \frac{a}{r+1} (\log x)^{r+1}, & r > -1, \\ a \log \log x, & r = -1. \end{cases} \quad (9.5)$$

Furthermore,  $\sum_{j \leq x} j^3 a_j = O(x(\log x)^r)$ .

*Proof.* For each fixed  $\epsilon > 0$ , the weights beyond a fixed index lie between  $(1 - \epsilon)$  and  $(1 + \epsilon)$  times their comparison sequence. Integral comparison, followed by  $\epsilon \downarrow 0$ , reduces the assertions to the corresponding integrals. For the tails, integration by parts shows that differentiating the logarithmic factor produces a relative  $O(1/\log x)$  error. For the variance, integrate  $(\log t)^r/t$ ; it has primitive  $(\log t)^{r+1}/(r+1)$  when  $r > -1$ , and  $\log \log t$  when  $r = -1$ . Finally,  $\int^x (\log t)^r dt = O(x(\log x)^r)$  for fixed  $r$ . Finite prefixes are negligible because  $V(x) \rightarrow \infty$ .  $\square$

**Theorem 9.2** (Log-weighted critical chart at leading order). *At  $\beta = \beta_c$ , the small positive critical inverse satisfies*

$$\varepsilon = x + \beta_c (A(e^{-x}) - d_0) \sim \frac{\beta_c}{2} x^2 V(1/x), \quad x \downarrow 0. \quad (9.6)$$

Consequently,

$$x \sim \begin{cases} \sqrt{\frac{2(r+1)\varepsilon}{\beta_c a}} \left( \frac{1}{2} \log \frac{1}{\varepsilon} \right)^{-(r+1)/2}, & r > -1, \\ \sqrt{\frac{2\varepsilon}{\beta_c a \log \log(1/\varepsilon)}}, & r = -1. \end{cases} \quad (9.7)$$

In both cases the critical inverse has no nonzero convergent finite Puiseux representation at  $\varepsilon = 0$ .

*Proof.* Criticality cancels the linear term exactly:

$$\varepsilon = \beta_c \sum_{j \geq 1} a_j (e^{-jx} - 1 + jx).$$

For  $j \leq 1/x$ , Taylor's formula differs from  $j^2 x^2/2$  by at most a constant times  $j^3 x^3$ . Lemma 9.1 bounds the sum of these errors by  $O(x^2(\log(1/x))^r)$ . For  $j > 1/x$ ,  $0 \leq e^{-jx} - 1 + jx \leq jx$ , so the tail has the same upper bound. Both errors are  $o(x^2 V(1/x))$  for every  $r \geq -1$ . This proves (9.6).

Taking logarithms first gives  $\log(1/x) \sim \frac{1}{2} \log(1/\varepsilon)$ . Substituting this into the slowly varying factors in (9.5) gives (9.7). In each case the logarithmic order of  $x$  is  $\varepsilon^{1/2}$  but  $x/\sqrt{\varepsilon} \rightarrow 0$ , excluding a nonzero convergent finite Puiseux series exactly as in Proposition 3.4.  $\square$

**Theorem 9.3** (Two scales and a universal leading cutoff law). *Let  $\beta = \beta_n$  lie in a compact positive interval, and suppose*

$$\lambda_n^* = \frac{n(1 - \beta_n d_1)}{b_n^*}$$

*lies in a compact real interval. Then, uniformly in that window,*

$$u_n(\beta_n) = \frac{\rho_{\beta_n}^{-n}}{nb_n^*} (\phi(\lambda_n^*) + o(1)). \quad (9.8)$$

*If  $M/e_n^* \rightarrow s \in (0, \infty)$ , then*

$$\frac{u_n^{(M)}(\beta_n)}{u_n(\beta_n)} \rightarrow e^{-1/(2s^2)}. \quad (9.9)$$

*More generally, the ratio tends to 1 if and only if  $M/e_n^* \rightarrow \infty$ , and tends to zero when  $M/e_n^* \rightarrow 0$ . The conditioned extreme-action point process at scale  $e_n^*$  has the same Poisson limit with intensity  $x^{-3} dx$  as in Theorem 7.1.*

*Proof.* The displayed scales satisfy

$$\frac{n\beta V(b_n^*)}{(b_n^*)^2} \rightarrow 1, \quad \frac{e_n^*}{b_n^*} \rightarrow 0, \quad n\beta \sum_{j > se_n^*} a_j \rightarrow \frac{1}{2s^2}. \quad (9.10)$$

Indeed,  $\log b_n^* \sim \log e_n^* \sim \frac{1}{2} \log n$ , and Lemma 9.1 applies. For  $r = -1$  the squared ratio  $(b_n^*/e_n^*)^2$  is asymptotic to  $(\frac{1}{2} \log n) \log \log n$ ; for  $r > -1$  it is asymptotic to  $\log n / (2(r+1))$ .

For fixed real  $s$ , split the centered characteristic exponent at  $j = b_n^*$ . The part with  $j \leq b_n^*$  has quadratic term  $-s^2 n\beta V(b_n^*) / (2(b_n^*)^2) \rightarrow -s^2/2$ . The absolute Taylor remainder is bounded by a constant times

$$\frac{n\beta}{(b_n^*)^3} \sum_{j \leq b_n^*} j^3 a_j = O\left(\frac{(\log b_n^*)^r}{V(b_n^*)}\right) = o(1).$$

For  $j > b_n^*$ , use  $|e^{iu} - 1 - iu| \leq 2 + |u|$  and the tail estimates; the bound has the same order. This proves the Gaussian characteristic limit.

The local limit follows by the same localization argument as before. At small  $\theta$ , summing over  $j \leq 1/|\theta|$  gives the lower bound  $C\theta^2 V(1/|\theta|)$  for the damping. On  $|s| \leq (b_n^*)^{1/2}$ , the ratio  $V(b_n^*/|s|)/V(b_n^*)$  is bounded below by a positive constant, using (9.5); hence there is Gaussian damping on that interval. On the next interval the exponent is bounded below by a positive multiple of  $b_n^*$ , after choosing a sufficiently small fixed angular endpoint. This follows from the eventual increase of  $\theta^2 V(1/\theta)$  for the explicit logarithmic comparison functions. Outside it,  $a_1 > 0$  supplies a fixed gap. Dominated Fourier inversion therefore gives an additive local limit, uniformly at all integer arguments. Equation (2.6) proves (9.8).

At  $M \sim se_n^*$ , the retained variance has the same leading size:  $V(M)/V(b_n^*) \rightarrow 1$ . The omitted mean on the  $b_n^*$  scale tends to zero, since

$$\frac{n\beta}{b_n^*} \sum_{j > M} j a_j = O(e_n^*/b_n^*) = o(1).$$

The third scaled cumulant of the truncated family tends to zero as well. Truncated damping, obtained by stopping the lower-bound sum at  $\min(M, 1/|\theta|)$ , proves its Gaussian local limit. The exact identity (2.7) and (9.10) then give (9.9). Monotonicity proves the budget criterion for arbitrary cutoffs, as in Theorem 6.1.

Finally, the Riemann sums at scale  $e_n^*$  satisfy  $n\beta \sum a_j f(j/e_n^*) \rightarrow \int f(x) x^{-3} dx$  for compactly supported continuous  $f$  away from zero. The relative tail asymptotic in (9.1) is uniform over

such index ranges once their lower endpoint tends to infinity. Since  $e_n^*/b_n^* \rightarrow 0$ , the Fourier argument of Theorem 7.1 applies unchanged. Its uniform local-limit bound also supplies the necessary far-tail control.  $\square$

*Remark 9.4* (What is not inherited by weakening the tail hypothesis). An asymptotic relation  $a_j \sim aj^{-3}(\log j)^r$  allows arbitrary slow convergence and oscillatory lower-order errors. It does not imply an analytic remainder, a convergent Lambert–analytic chart, or the rate in Theorem 6.3. Theorems 9.2 and 9.3 deliberately make only leading-order assertions. The case  $r = -1$  also shows that simply replacing a divergent variance by a single  $\log n$  is not valid throughout the slowly varying class.

## 10 Reproducible computation and numerical diagnostics

### 10.1 Exact checks and numerical method

The accompanying program `code/verify.py` is a standalone reproduction script. Its exact rational/symbolic checks verify the Lambert chart equation through  $T^2$ , the finite-fold series  $Y$  through  $z^7$ , and the Lagrange coefficient identity through degree 6 in a rational test model. The additional drift and curvature series are checked by exact substitution. These finite checks detect algebraic mistakes; they are not substitutes for the all-order proofs.

For numerical diagnostics we use the pure model  $a = 1$ ,  $P = 0$ , so

$$d_0 = \zeta(3), \quad d_1 = \zeta(2), \quad \beta_c = 1/\zeta(2), \quad h = 3/2.$$

The script computes normalized coefficients with a discrete Fourier coefficient extraction at radius  $r = e^{-3/n}$ , using a power-of-two period  $L \geq 16(n+1)$ . Terms in  $A$  above degree  $n$  can be discarded *exactly* when extracting the coefficient of degree  $n$ . The normalized exponential has nonnegative coefficients with total mass at most 1. Therefore its coefficient aliases, in exact arithmetic, are nonnegative and have total error at most  $r^L = e^{-3L/n}$  after dividing by  $r^n$ . Floating-point roundoff is separate from that alias bound and is not rigorously enclosed.

An independent 80-decimal-digit coefficient recurrence at  $n = 128$  agreed with the FFT value to a relative discrepancy of about  $6.75 \times 10^{-15}$ . The recurrence is

$$p_0 = e^{-n\beta d_0}, \quad p_k = \frac{n\beta}{k} \sum_{j=1}^{\min(k,M)} j a_j p_{k-j}, \quad (10.1)$$

with  $M = n$  for the full coefficient. No conclusion about global floating-point certification follows from this one comparison.

### 10.2 Sector coefficients: inverse logarithms converge slowly

Write  $p_n = \mathbb{P}(S_{n,\beta_c} = n)$ . Table 1 compares the normalized mass with the first two corrections in Corollary 5.3. Even at  $n = 65536$ ,  $\ell_n$  is only about 7.825. The two-correction approximation is closer in these examples, but convergence is visibly slow. This behavior is consistent with an inverse-logarithmic expansion and warns against estimating its accuracy from the size of  $n$  alone.

### 10.3 Cutoff and critical-window diagnostics

Table 2 uses  $M$  nearest to  $e_n$ ; the effective value  $s_n = M/e_n$  is shown to avoid silently replacing an integer cutoff by its continuous target. The refined prediction exponentiates the explicit

| $n$   | $\ell_n$ | $b_n p_n / \phi(0)$ | $1 + g_1 / \ell_n$ | $1 + g_1 / \ell_n + g_2 / \ell_n^2$ |
|-------|----------|---------------------|--------------------|-------------------------------------|
| 256   | 4.8090   | 0.98892962          | 1.03793096         | 1.00781273                          |
| 1024  | 5.5761   | 0.99803178          | 1.03271250         | 1.01031140                          |
| 4096  | 6.3329   | 1.00306096          | 1.02880338         | 1.01143623                          |
| 16384 | 7.0820   | 1.00592159          | 1.02575691         | 1.01186924                          |
| 65536 | 7.8250   | 1.00756641          | 1.02331113         | 1.01193568                          |

Table 1: Critical sector diagnostics for the pure cubic-tail model. The last two columns are asymptotic approximations, not upper bounds.

| $n$   | $M$ | $s_n$  | Actual ratio | Leading   | Refined   |
|-------|-----|--------|--------------|-----------|-----------|
| 256   | 12  | 0.9619 | 0.6241867    | 0.5825271 | 0.6209904 |
| 1024  | 25  | 1.0020 | 0.6424435    | 0.6077365 | 0.6395692 |
| 4096  | 50  | 1.0020 | 0.6371199    | 0.6077365 | 0.6347475 |
| 16384 | 100 | 1.0020 | 0.6344374    | 0.6077365 | 0.6324177 |
| 65536 | 200 | 1.0020 | 0.6329721    | 0.6077365 | 0.6312248 |

Table 2: Retained coefficient fractions near the critical action budget. The refined prediction incorporates omitted mass, mean, and variance.

part of (6.7), retaining the exact finite sums  $\nu_{n,M}$ ,  $\kappa_{n,M}$ , and  $v_{n,M}$ . The leading prediction approaches the correct limit but misses substantial finite- $n$  corrections. The refined formula reduces those discrepancies in the displayed range; the remaining differences are not treated as certified error intervals.

Table 3 changes  $\beta$  at  $n = 4096$  to obtain  $\lambda_n = -1, 0, 1$ . The density ratio is  $b_n \mathbb{P}(S = n) / \phi(\lambda_n)$ , its first correction is  $1 + g(\lambda_n) / \ell_n$ , and the cutoff is again nearest to  $e_n$ . The cutoff's leading limit is independent of  $\lambda_n$ , but its finite-window correction is not.

| $\lambda_n$ | Density ratio | First correction | Cutoff ratio | Refined cutoff |
|-------------|---------------|------------------|--------------|----------------|
| -1          | 1.1651828     | 1.1303772        | 0.8218048    | 0.8143599      |
| 0           | 1.0030610     | 1.0288034        | 0.6371199    | 0.6347475      |
| 1           | 0.8072274     | 0.8039156        | 0.3716506    | 0.3690131      |

Table 3: Finite-coupling diagnostics at sector index 4096.

## 10.4 Inverse chart and moving folds

In the same pure model, the relative errors of the core  $T$  and its first two analytic corrections are as follows:

| $\varepsilon$ | Core $T$                | $T + v_1 T^2$           | $T + v_1 T^2 + v_2 T^3$ |
|---------------|-------------------------|-------------------------|-------------------------|
| $10^{-4}$     | $1.0020 \cdot 10^{-4}$  | $5.5798 \cdot 10^{-8}$  | $2.7144 \cdot 10^{-11}$ |
| $10^{-8}$     | $4.1471 \cdot 10^{-7}$  | $1.3691 \cdot 10^{-12}$ | $2.9478 \cdot 10^{-18}$ |
| $10^{-16}$    | $1.6274 \cdot 10^{-11}$ | $3.3625 \cdot 10^{-21}$ | $3.0045 \cdot 10^{-31}$ |

The reference roots were computed at 80-decimal-digit working precision, not enclosed by interval arithmetic. Here the exact Lambert core has already absorbed the logarithmic singularity, and the remaining analytic expansion is correspondingly effective.

For finite folds, Table 4 compares the exact numerical critical point and value with (8.18) and (8.19). The radius column tends to 1, while the three-term prediction includes the  $L_M^{-1}$  and  $L_M^{-3}$  terms. Radius drift was evaluated using the cancellation-resistant identity (8.14).

| $M$  | $ML_M \log \tau_M$ | $2M^2 \log(r_M/\rho_c)/c$ | Three-term prediction |
|------|--------------------|---------------------------|-----------------------|
| 32   | 0.95564674         | 1.20338010                | 1.24141038            |
| 128  | 0.97935090         | 1.17279266                | 1.18197701            |
| 512  | 0.98825384         | 1.14341348                | 1.14565007            |
| 2048 | 0.99228716         | 1.12076653                | 1.12131621            |
| 8192 | 0.99446566         | 1.10378025                | 1.10391682            |

Table 4: Finite-action critical-point and critical-value drift. In this pure model  $L_M = H_M$ .

## 10.5 Reproduction and reliability boundary

From the package directory, run

```
python -m pip install -r requirements.txt
python code/verify.py --max-n 65536
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A third LaTeX pass may be used to stabilize references. The generated TeX tables are included, so the PDF can be rebuilt without rerunning the numerical code. The script resolves output paths relative to its own package root, not the caller's working directory. It overwrites only its own generated data files.

The full machine-readable record is `data/verification.json`; the supplied console output is `data/verification_run.txt`. Exact checks, numerical diagnostics, analytic theorems, and prospective formalization are distinct layers. The program does not prove Fourier dominating estimates by sampling, and it does not verify analytic continuation or quantify a theorem's unstated constants.

## 11 Consequences for transseries and proof architecture

### 11.1 Three expansions, three different meanings

There are three variables and three limiting processes. Near  $q = 0$ ,  $U_\beta(q)$  is an ordinary convergent series; after  $q = e^{-X}$  it is an integer-action exponential transseries. Near the critical value  $q = \rho_c$ , the local variable is  $\varepsilon = \log(\rho_c/q)$ , and the inverse uses the convergent Lambert chart  $x = TV(T, \eta)$ . Finally, large sector index  $n$  has an asymptotic expansion in  $1/\ell_n$ , with exponentially weighted prefactor  $\rho_\beta^{-n}$ .

Convergence of the small- $q$  germ says nothing by itself about analytic ramification at  $\rho_c$ . Convergence of  $V$  in two independent variables does not imply convergence of the large- $n$  inverse-logarithmic series. The two uses of Lambert  $W$  are related by the same logarithmic balance, but their arguments have different roles: one inverts the critical map, and the other normalizes a Fourier saddle.

### 11.2 What finite-prefix universality actually preserves

Changing finitely many  $a_j$  can change  $d_0$ ,  $d_1$ , and  $h$ . Consequently it changes the critical coupling, the exponential sector factor, the exact normalization, and the analytic coefficients of the critical inverse. It does not change the universal functions  $G_j$  after that exact normalization, the leading cutoff profile, or the theorems' logarithmic orders. In the finite-fold chart it is

absorbed in  $L_M = H_M + p^{[2]}/a$ , up to the displayed algebraically small errors. These are precise invariance statements, not a claim that different finite prefixes produce identical functions.

The leading sector expansion also cannot detect every piece of the analytic remainder. At every fixed inverse-logarithmic order, the  $n/b_n^3$  contribution is smaller than the error scale; it becomes relevant in a finer multi-scale expansion. This supplies a natural next level of the transseries rather than a reason to discard the analytic remainder.

### 11.3 A feasible formalization decomposition

The repository's existing asymptotic-scale and residual-transport work suggests a modular proof path, but no file in this package is claimed to be machine-checked. A practical decomposition would be:

1. Formalize the exact coefficient identities and the finite-series recurrences first. These are algebraic and can be separated from analytic convergence and probability.
2. Formalize the Lambert core identity and the local implicit-chart interface with explicit hypotheses for holomorphicity, branch choice, and nonzero derivative. The geometric Cauchy bound is then a reusable separate theorem.
3. Formalize the finite-fold rescaling identities and Riemann-sum error estimates, keeping exact critical quantities separate from asymptotic approximations. Proposition 8.4 is a particularly useful safety interface.
4. Package Fourier inversion, the damping lemmas, and an integrable-majorant expansion theorem. The uniform-in- $k$  additive bound should be represented explicitly, rather than inferred from pointwise convergence.

The conditioned Poisson process and slowly varying extension could then be built on those analytic foundations. A declaration-level crosswalk to existing Lean files would require additional source inspection and actual compilation; neither is asserted here.

## 12 Further research questions and concrete targets

**1. A genuine two-parameter upper-endpoint crossover.** Let  $a_j = aj^{-1-\alpha_n}$  with  $\alpha_n \uparrow 2$ . Determine the joint limit when  $(2 - \alpha_n) \log n$  stays bounded. The truncated variance involves the cancellation in  $(b^{2-\alpha} - 1)/(2 - \alpha)$ ; this suggests a transition parameter, but neither the critical chart nor the cutoff kernel follows by substitution. The target is a uniform chart and a uniform coefficient law connecting the fractional stable regime to this fixed endpoint.

*Editorial note (ProveIt, 2026-09-29).* Answered, pending review, by two later packages of this series written independently of each other. The stable-Gaussian package [11] works in exactly this window,  $|2 - \alpha_n| \log n \leq K$ , from both sides, and gives a two-term uniform inverse, a convergent chart about a desingularized core, the uniform coefficient law with its first correction, and a uniform first cutoff correction without finite-prefix moments; it did not see this article. The confluent package [10] allows  $2 - \alpha_n \rightarrow 0$  at any rate and of either sign, with one convergent chart in three variables and a leading-order coefficient law; it read this article but does not cite this question by number. Where both apply they agree term by term, and at  $\alpha = 2$  the stable-Gaussian first cutoff correction is (6.13), up to the change of logarithmic normalization.

**2. Explicit finite- $n$  constants for action selection.** Turn Theorem 6.3 into an inequality with numerical constants, threshold values, and interval-enclosed  $G_1$ . Given  $n$ ,  $\beta$ , and a relative

loss tolerance, return the smallest certified cutoff or a certified near-minimal interval. Formula (6.14) is an asymptotic design rule, not that algorithm.

*Editorial note (ProveIt, 2026-09-29).* Partial progress: for the same model the marginal package [9] proves an explicit finite- $n$  lower bound on the retained fraction (its `thm:finite-certificate`) and a two-term minimal budget (its `thm:min-budget`). Its recorded evaluations use no directed rounding, and the certified selection algorithm remains open there too.

**3. Large orders of the Gaussian–logarithmic kernels.** Determine the growth of  $G_j(\lambda)$  as  $j \rightarrow \infty$ , uniformly on specified real or complex  $\lambda$  sets. Does the inverse-logarithmic series admit a useful Borel or accelerated summation? Identify an optimal truncation index and compare its remainder with the first analytic  $n^{-1/2}$ -scale corrections. No convergence assertion for this series is proved above.

**4. The finer transseries beyond every inverse-logarithmic order.** Include the analytic terms of  $R_\beta$  in (5.8) and construct a joint expansion in  $1/\ell_n$  and  $n/b_n^3$ . The leading new Fourier factor is cubic and depends on  $r_0$ . A desirable result is an all-order multi-index formula with an error estimate that does not double-count corrections absorbed in the exact Lambert scale.

**5. An expanding critical coupling window.** Find the largest growing range of  $\lambda_n$  on which a finite correction to (5.3) is relatively accurate. The present theorem is uniform additively for all integers but only yields a useful relative law on compact windows. Match large positive and negative  $\lambda_n$  to inherited-tail and interior-saddle regimes, respectively.

**6. Beyond logarithmic powers in the slowly varying tail.** For  $a_j \sim aj^{-3}\mathcal{L}(j)$  with a general slowly varying  $\mathcal{L}$ , characterize conditions under which the extreme and fluctuation scales separate and conditioning remains asymptotically invisible to extremes. Specify an effective inverse notation when the normalization requires a de Bruijn conjugate rather than Lambert  $W$ . The explicit log-power subclass does not resolve this general problem.

**7. Finite- $M$  corrections to the drift chart.** Augment Theorem 8.1 by powers of  $1/M$  with coefficient functions analytic in  $1/L_M$ . The tail sums and Riemann sums have Euler–Maclaurin expansions, but one must keep their finite-prefix terms and cancellations synchronized. The target is an exponentiable formula in regimes where  $n/M^3$  no longer tends to zero.

**8. The lower endpoint and loss of the mean.** At a tail near  $j^{-2}$ ,  $d_1$  may diverge and the critical coupling  $1/d_1$  collapses. Determine an appropriate renormalized coupling, critical coordinate, and action budget. This is not the same infinite-variance Gaussian problem and needs a new normalization.

**9. Nonlinear slopes and genuinely noncommensurate actions.** Replace  $jU$  by  $(j+r_j)U$ , or replace integer actions by a sequence with no common lattice. Determine which endpoint and cutoff laws persist. The exact substitution  $t = qe^U$  or lattice Fourier inversion then fails, so a stability theorem must replace the exact formulas.

**10. Signed weights and competing complex critical values.** Allow controlled signed or complex perturbations of the action tail. Find analytic hypotheses replacing positivity and Poisson conditioning, and identify when multiple complex saddles contribute. Neither the monotone cutoff criterion nor its probabilistic proof extends unchanged to that setting.

**11. Minimal transseries support and monodromy of the Lambert chart.** Determine a minimal convenient well-based scale for expanding  $TV(T, \eta)$  after replacing  $W_{-1}$  by its nested-logarithmic expansion. Describe analytic continuation between its logarithmic branches, with domain bounds. The no-Puiseux proposition rules out only finite algebraic ramification; it does not classify all Hahn or logarithmic representations.

**12. Proof-producing computation.** Build a checker for the exact chart recurrences, the finite-fold stationarity cancellation, and explicit local holomorphic majorants. Then add interval Fourier quadrature with a proved tail enclosure. Connect these certificates to formal statements distinguishing exact coefficient identities, analytic existence, and numerical enclosures.

## 13 Conclusion

The upper endpoint of the positive power-tail action model is governed by a logarithmic critical singularity. Its local inverse becomes an analytic convergent function after an exact Lambert change of scale. Its large-sector coefficients have an independently derived, arbitrary-order inverse-logarithmic Gaussian expansion. The action cutoff is controlled by a smaller extreme scale, and its first finite-window corrections retain information that the leading Poisson profile suppresses.

The finite-action folds provide a second, analytic explanation of the same cutoff phenomenon. Their drift and curvature are controlled to all inverse-logarithmic orders by convergent implicit functions, with separate algebraic errors that must be multiplied by  $n$  before exponentiation. The log-weighted extension demonstrates that the scale separation persists beyond the exact polylogarithm, while also making clear which convergence and rate claims depend on the stronger tail hypothesis.

These results resolve the fixed-endpoint problem and several defined extensions within the stated model. They do not settle the full joint exponent crossover, general resurgence, arbitrary slowly varying realization, or formal verification. Those remaining targets are now separated from the statements proved here.

## A Source audit, dependencies, and research status

### A.1 Repository snapshot and inspected scope

The repository was accessed at the pinned commit

db68f0853c3c69cd930caedab6f6ad9ad9adb11eaf.

The focused inspection included the directory tree under `Analysis/Transseries`, the group README under `docs/series-and-transseries`, and the moving-fold package’s README. The group README distinguishes consolidated material from unmerged research arrivals and records formalization limitations. This article does not convert those human manuscript claims into machine-checked results.

The preceding countable-action article [8] was inspected as its editable TeX in the user’s Library, named `article(20260929-193550).tex`. Its model, principal theorem statements, and research-question section were read. In particular, its Section 11, Question 4 asks for endpoint logarithmic charts and cutoff laws, and Questions 1, 2, and 7 motivate the extensions above. Its presence in the Library does not establish that it was merged into the pinned GitHub snapshot. This provenance is kept separate from the repository snapshot.

This was a targeted research audit, not a line-by-line review of the entire canonical transseries volume, all recent articles, or the Lean corpus. The proofs in the present article do not assume the correctness of a new theorem in an unreviewed repository package.

## A.2 External sources and priority boundary

The primary sources consulted were the NIST DLMF entries for polylogarithms and Lambert  $W$ , the author-maintained site for *Analytic Combinatorics*, Janson’s article (including its Example 18.29 in the arXiv HTML version), and the Chakrabarty–Samorodnitsky truncation paper’s bibliographic record. The classical leading Gaussian/extreme mechanism is explicitly acknowledged in Section 1.

The specialized formulas and proofs assembled here warrant independent mathematical review. A literature search finding no exact counterpart would still not establish priority by itself. Accordingly, this package uses “contribution developed here” rather than an unsupported claim that a longstanding worldwide conjecture has been resolved.

## A.3 Proof dependencies

The model identities in Section 2 feed two independent routes. The local analytic route uses the integer-order polylogarithm expansion and the implicit function theorem to obtain the Lambert chart. The coefficient route uses the exact Poisson identity, Fourier inversion, and the damping lemma to obtain all-order sector estimates. The refined cutoff theorem combines that full local limit with a separate quantitative truncated local limit. The moving-fold route is based on exact critical-point identities and controlled Riemann sums, and only then uses a tilted local limit for coefficients. The log-weighted extension uses its own leading tail estimates and does not invoke an analytic remainder it has not established.

## B Notation ledger

| Symbol                             | Meaning                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------|
| $A, A_M$                           | Full action generating function and its cutoff at action $M$ .                        |
| $a_j, a, P$                        | Nonnegative action weights, cubic-tail amplitude, finite-prefix polynomial.           |
| $p^{[k]}$                          | Finite-prefix moment $\sum_{j \leq J} j^k p_j$ , not a full moment.                   |
| $d_0, d_1$                         | Finite endpoint mass and mean: $A(1)$ and $\sum j a_j$ .                              |
| $\beta_c, \rho_\beta, c_\beta$     | Critical coupling $d_1^{-1}$ , factor $e^{-\beta d_0}$ , and $\beta a$ .              |
| $h$                                | Exact quadratic normal-form constant $3/2 + p^{[2]}/a$ .                              |
| $U_\beta, u_n$                     | Feedback solution and its coefficient of $q^n$ .                                      |
| $t, x, \varepsilon$                | $qe^U$ , $-\log t$ , and $\log(\rho_\beta/q)$ .                                       |
| $R_\beta, r_k$                     | Analytic normal-form remainder and its critical Taylor coefficients.                  |
| $T, \eta, V$                       | Exact Lambert core, reciprocal logarithmic coordinate, and convergent analytic chart. |
| $Z_j, S, S_M$                      | Independent Poisson action counts and their full/truncated weighted sums.             |
| $e_n, b_n, \ell_n$                 | Extreme scale $\sqrt{n\beta a}$ , fluctuation scale, and exact Lambert logarithm.     |
| $\lambda_n, s_n$                   | Critical-window coordinate $n(1 - \beta d_1)/b_n$ and cutoff ratio $M/e_n$ .          |
| $G_j, g_j, g$                      | Fourier correction kernels, their normalized values at 0, and $G_1/\phi$ .            |
| $R_{n,M}$                          | Retained coefficient fraction $u_n^{(M)}/u_n$ .                                       |
| $\nu_{n,M}, \kappa_{n,M}, v_{n,M}$ | Omitted intensity, omitted scaled mean, and retained scaled variance.                 |
| $\tau_M, r_M, \mathcal{V}_M$       | Exact finite-fold critical point, critical value, and tilted curvature.               |
| $\sigma_M, y_M, L_M$               | $\log \tau_M$ , $M \log \tau_M$ , and $H_M + p^{[2]}/a$ .                             |

| Symbol                    | Meaning                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| $F, G, J, Y, \mathcal{H}$ | Entire or local analytic functions describing the finite-fold drift. Here $G$ without a subscript is not the Fourier kernel $G_j$ . |
| $e_n^*, b_n^*, V(x)$      | Leading scales and truncated second moment for log-weighted tails.                                                                  |

## References

- [1] NIST Digital Library of Mathematical Functions. *Section 25.12: Polylogarithms*, especially Eq. 25.12.12 and its restriction on the order parameter. <https://dlmf.nist.gov/25.12>. Consulted 29 September 2026.
- [2] NIST Digital Library of Mathematical Functions. *Section 4.13: Lambert W-Function*. <https://dlmf.nist.gov/4.13>. Consulted 29 September 2026.
- [3] P. Flajolet and R. Sedgewick. *Analytic Combinatorics*. Cambridge University Press, 2009. Author-maintained book site: <https://ac.cs.princeton.edu/home/>.
- [4] S. Janson. Simply generated trees, conditioned Galton–Watson trees, random allocations and condensation. *Probability Surveys* **9** (2012), 103–252. doi:10.1214/11-PS188. arXiv:1112.0510. The cubic-tail comparison discussed here is Example 18.29 in the inspected arXiv HTML version: <https://arxiv.org/html/1112.0510v1>.
- [5] A. Chakrabarty and G. Samorodnitsky. Understanding heavy tails in a bounded world or, is a truncated heavy tail heavy or not? *Stochastic Models* **28**(1) (2012), 109–143. doi:10.1080/15326349.2012.646551. arXiv:1001.3218.
- [6] VladimirReshetnikov/ProveIt. *Series and transseries*, group README and package inventory. Pinned commit and audit boundary: Appendix A. <https://github.com/VladimirReshetnikov/ProveIt/tree/db68f0853c3c69cd930caedab6f6ad9addb11eaf/Analysis/Transseries/docs/series-and-transseries>.
- [7] ProveIt repository, research package. *Critical Transseries at a Moving Fold: A uniform two-sheet atlas, large-sector crossover, and resonant action splitting*. September 2026. Package README inspected at the pinned commit, subdirectory `Critical_Transseries_Moving_Fold` of [6].
- [8] *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*. Earlier research draft prepared for Vladimir Reshetnikov, 29 September 2026. Editable source inspected in the user’s Library as `article(20260929-193550).tex`; see especially Section 11, Questions 1, 2, 4, and 7. Repository inclusion at the pinned snapshot is not assumed. Editorial note (ProveIt, 2026-09-29): the article is now filed as `Analysis/Transseries/docs/series-and-transseries/Critical_Hahn_Transseries_Beyond_Finite_Action_Folds/article.tex` (batch 47).
- [9] [Editorial addition, ProveIt, 2026-09-29.] *Marginal Critical Transseries: Convergent Lambert-W Charts, Universal Cutoff Corrections, and Certified Action Budgets*, research package filed in batch 48 in the group of [6]: `Marginal_Critical_Transseries_Action_Budgets/Marginal_Critical_Transseries.tex`.
- [10] [Editorial addition, ProveIt, 2026-09-29.] *Confluent Critical Transseries Across the Exponent-Two Boundary*, research package filed in batch 49 in the group of [6]: `Confluent_Critical_Transseries_Exponent_Two_Boundary/Confluent_Critical_Transseries.tex`.

- [11] [Editorial addition, ProveIt, 2026-09-29.] *Through the Stable-Gaussian Endpoint: Uniform Critical Transseries, Prefix-Independent Cutoff Corrections, and Conditional Extremes*, research package filed in batch 49 in the group of [6]: `Stable_Gaussian_Endpoint_Uniform_Critical_Transseries/article.tex`.